ENGINEERING REFERENCE / VERSION 04

Halo 2
Original Matchmaking
Parity Master Reference

Algorithms • Historical profiles • Validation evidence

Prepared for Eric Krouss | October 28, 2026

Required outcome: original-mechanism parity across all original-valid multiplayer modes, variants and historical playlist profiles.

This reference separates recovered logic, source-traced contracts, reported original controls, locally executed documentation checks, modern replacements and unresolved historical evidence. It is a reconstruction specification, not a blanket certification of the current port.

Release disposition: RESEARCH AND IMPLEMENTATION REFERENCE. COMPLETE ORIGINAL-SERVICE PARITY IS NOT YET CERTIFIED.

The audited native snapshot is `ec3fdb77161886514b67bcdfd6bc7ec5d6659c88`. The independently inspected decompilation snapshot is `ff5f319fcdfc8c0b4ea7879e4bcebd69a384ec8e`. These are separate repositories and authorities. V4 does not modify either repository or a deployment.

Reading map and package contract

Start with Sections 1 and 2 for the goal and evidence boundaries. Sections 3–6 describe rank, search, teams and game selection. Sections 7–10 connect all modes, historical catalogs, session continuity and modern service responsibilities. Sections 11–13 define validation, the original algorithm registry and the remaining closure work. Section 14 identifies sources.

The machine-readable registry is part of the specification, not an optional attachment. It records 53 algorithm/ownership entries with original addresses where established, address kinds, inputs, outputs, source pins, native owners, test references and unresolved gaps. An unknown address or unrun test is represented as unknown, not inferred from a neighboring version.

Package item	Purpose
This master reference, Word and Markdown	Human-readable contracts, formulas, findings and release requirements.
`registry/original_algorithm_registry_v4.json`	Per-owner input/output and evidence contracts.
`registry/parity_coverage_register_v4.json`	Seven mode families, 39 built-in presets, historical profile checkpoints and retained V3 requirements.
`registry/gap_closure_register_v4.json`	Named blockers and evidence required to close each one.
`data/retail_fallback_pool_rows_v4.json`	61 literal source rows and selector mapping; not a deployable historical catalog.
`data/rank_contract_inputs_v4.json`	Source-transcribed explanatory rank inputs for documentation tests.
`tests/`	Executed contract checks, retained counterexample, results and fail-closed release validator.
`evidence/`	Source manifest, retrieval log, research scope and retained prior coverage plan.
`handoff/CODEX_IMPLEMENTATION_CONTRACT_v4.md`	Mandatory implementation and evidence-maintenance workflow.

1. Final delivery goal and non-negotiable scope

1.1 What “the same mechanism” means

For equal valid original inputs, executable/configuration epoch, initial state, event order and random state, the compatibility core must produce the same observable decisions as the original reference. The required outputs include eligibility, query envelopes, candidate ordering and selection, party/team assignment, selected map and complete game variant, host/session transitions, final standings and player/clan XP and rank changes.

This is not a promise of identical historical opponents or queue times. Today's population, routes and platforms are different. It is a testable equivalence requirement for the decision mechanism and the declared input contract. A new backend can implement that contract, but its own undocumented session ordering, QoS substitutions or result policy must not be passed off as the recovered Xbox Live implementation.

The program is not complete when one playlist works, when the rank formula matches, or when all 39 menu presets load. Completion requires every entry in the reconciled historical scope to pass its applicable original-reference and end-to-end gates. A narrower interim profile may ship with an honest scope label; it does not erase the full restoration goal.

1.2 Mode, variant, playlist and epoch are different objects

A mode is an engine family. A variant is an authored settings buffer. A playlist or hopper selects eligible players and a pool of map/variant entries. A historical revision supplies a dated set of rules and settings. One playlist can use several modes, and one mode can appear in many playlists. Neither a UI name nor a current playlist label is sufficient to identify the full configuration.

Coverage includes Slayer, Capture the Flag, Assault, Oddball, King of the Hill, Territories and Juggernaut. It includes ranked and hidden-skill casual queues, solo and party admission, original-valid two-team/multi-team layouts, large sessions, local accounts/guests, clan lineups, departures, host changes, postgame results and party return. It does not invent unsupported combinations by taking the Cartesian product of all maps, modes and team counts. [S17; N07–N10]

Custom, LAN and offline multiplayer must not write matchmaking XP. Campaign is not an original competitive hopper. Modern custom modes, including explicitly native offerings, need separate configuration and claims. [S17; V3 requirements]

1.3 Historical breadth without a fictional hybrid

The preferred late-classic experience and the historical-restoration program remain distinct. Preserve retail defaults, TU1.5 defaults, dated Bungie catalogs and genuinely recovered service files as separate profiles. An older restored Action Sack or clan playlist can be selectable without pretending it belonged to the final service catalog.

A profile must bind executable identity, rank policy, hopper data, complete map/variant rows, statistics namespace and evidence. Rename, visibility change, configuration change and rank reset are separate operations. Exposing a hidden rank does not by itself authorize resetting it. Reusing a familiar name does not authorize reusing a different ladder. [S09, S13, S19]

1.4 Modern improvements stay in scope

Steam-based identity, opaque internal player IDs, secure authorization, protected relay transport, reliable storage, idempotent requests and observability remain required modern engineering choices. Preserve one-clan membership. Do not give clients authority over persistent XP, rank, party membership, clan identity or measured route quality.

Security improvements are not a reason to reproduce exploitable trust boundaries. Equally, they are not permission to silently replace ranking, team assignment or playlist behavior. Record which improvements change admission, measurements, host choice or result finalization and keep those effects separate from original-kernel equivalence. [V3 project requirements; S11, S12; N10]

2. Evidence model and substantive V4 findings

2.1 What V4 actually did

V4 read Version 3, inspected pinned native code and a separately pinned decompilation, traced the final team-assignment caller and hopper getters, inspected the weighted selector and RNG helper, checked newer seven-family implementation evidence, read the actual result validator and pursued historical playlist/settings records. It created and ran source-derived numerical/data checks and reran the retained isolated two-team counterexample.

V4 did not execute or independently hash an original XBE, compile the native Rust kernel, run the complete game/server, inspect a live deployment, recover the original Xbox Live stored procedure or retrieve final-2010 network/hopper configuration files. The April 2006 spreadsheet announcement was found, but the spreadsheet itself could not be retrieved. These limits are part of the evidence record, not hidden footnotes.

2.2 Evidence labels

Label	Meaning and permitted claim
INSPECTED SOURCE	Code/data was actually read at a pinned revision. Supports statements about that source, not universal binary equivalence.
SOURCE-TRACED CONTRACT	A producer/caller/consumer path was followed in reconstructed code. Numeric fields and control flow still require original-byte validation where not matched.
RECORDED ORIGINAL CONTROL	A project artifact reports original execution. Attribute the result to that record; do not call it a V4 rerun.
EXECUTED CONTRACT CHECK	This package's reference/data test ran. It checks the supplied contract, not the original executable.
ORIGINAL/NATIVE REPLAY	Both the pinned original and target implementation actually ran with equivalent inputs and recorded outputs. Not performed in V4.
HISTORICAL INPUT VERIFIED	A dated original service input was recovered and authenticated. A public name/settings list may support only a subset of fields.
MODERN POLICY	A deliberately new service or security rule, not recovered historical behavior.

A decompilation address is not a matching-status certificate. The independently inspected upstream inventory still marks retail `0x71730` as `todo`. That matters because it is the caller controlling final assignment, even though much of the kernel is well documented. V4 records the distinction rather than upgrading the entire path to “exact.” [N01, N05]

2.3 What has advanced since V3

Area	New concrete result	Remaining boundary
Team assignment	Traced original hopper-derived flags, count feasibility, raw weights and conditional individual fallback.	Original caller replay and native integration remain required.
Balancing source	Inspected the independent greedy/local-swap source alongside the native Rust kernel.	Source agreement is not an independent XBE run.
Game selection	Identified LCG step, high-16-bit scaling, cumulative boundary and signed-short range issue.	Seed/call ownership, overflow cases and full settings need original controls.

Area	New concrete result	Remaining boundary
Fallback pool data	Packaged all 61 inspected literal rows, preserving duplicate rows and recorded 116 active instances.	These are not final-service files or complete authored buffers.
All-mode coverage	Mapped seven engine types and all 39 named built-ins; separated bounded mode tests from full online acceptance.	Historical downloaded variants extend beyond this set.
Results	Inspected all-owner confirmation and the known-name versus unknown-name validation branch.	Original arbitration and full variant-binding chain remain unresolved.
Historical scope	Added deployed Team Carnage/HBO evidence and the April 2006 settings-spreadsheet lead.	Full weights, buffers and final-2010 tuning are not supplied by those records.

2.4 Pinned identities

Artifact	Recorded identity
Native source	`ec3fdb77161886514b67bcdfd6bc7ec5d6659c88`
Independent decompilation	`ff5f319fcdfc8c0b4ea7879e4bcebd69a384ec8e`
Retail XBE SHA-256, project-recorded	`03215919bb7163259257d361f4c7bf802a7ab12aa85e2689436369b5c427935d`
TU1.5 English XBE SHA-256, project-recorded	`21b207e7efdacaf6124a0822837f8be570766a42505057f7fd2d2ba8f409c4fd`
Native retail rank JSON SHA-256 enforced by inspected source	`5b5760a487d6af59da885edb89f2d7d7e76df5fa0d2379cb7a43ff9dfa3e55bd`
Native TU1.5 rank JSON SHA-256 enforced by inspected source	`9ad56e3c3ef7249e8ada858a29016b4ecaca0148b9deb4efe87efa7ded15c58f`

The XBE hashes were read from project evidence, not recomputed from original binaries in this session. The explanatory JSON packaged with V4 has its own provenance and must not be relabeled as either original policy file. [S02, S09, S13]

3. Ranking and progression: complete recovered contract

3.1 Persistent state and comparison state

Store persistent XP, persistent raw rank, previous raw rank, effective pair-comparison skill, visible badge and statistics namespace separately. Internal raw rank is 0–127, while public presentation groups the final range at level 50. Search and arithmetic must not consume a capped UI value in place of the original raw state. [S02, S08, S13]

For each result, the original producer chain must establish effective skill. The existence of a party floor does not prove that every remote-party search, team assignment and XP path uses the same adjusted number. In the traced final-team caller, raw summary weights are passed to the balancing kernel. In the rank evidence, comparison skill and prior persistent rank are distinct fields. Preserve those boundaries and verify remote parties, multiple local accounts, guests and clan subjects independently. [N01; S08, S13]

3.2 Pair exchange

Retail pair routine: `0x746B0`. Recorded TU1.5 counterpart: `0x784D0`. Lower final standing is better. The no-standing sentinel is `-1`. A valid standing beats an absent standing; two equal standings, including two absent standings, contribute zero. [S09, S13]

```
index = own_effective_raw_skill - opponent_effective_raw_skill + 127

if own standing wins: use win exchange / own win factor / own win offset
if own standing loses: use loss exchange / own loss factor / own loss offset
if equal standings: return 0

pair_delta = trunc_toward_zero(exchange[index] * factor[own_skill] / 100)
             + offset[own_skill]
```

The recovered storage contract uses signed 16-bit exchange and offset values and unsigned 8-bit factors. Signed division truncates toward zero. Do not use Python floor division on a negative sum, round to nearest, or replace the lookup with Elo/TrueSkill. All default additive offsets described here are zero, but the general engine must retain them as configuration data. [S02, S13]

3.3 Aggregation and legal XP

Retail player aggregation is `0x74800`; clan aggregation is `0x74720`. Recorded TU1.5 counterparts are `0x78620` and `0x78540`. The player path compares the selected participant against eligible opposing participants, excluding self, same-team players and records rejected by the original statistics key, active/skill/team/identity checks. An eligible tied opponent still counts in the divisor. Team games compare final team standings, not personal kill/death order. [S02, S09, S13]

```
raw_delta = 0 when there are no eligible opponents
otherwise raw_delta = trunc_toward_zero(sum(pair_deltas) / opponent_count)

first constrain XP + delta to [0, 0x3fffffff]
then constrain delta to the configured minimum and maximum
new_XP = old_XP + delta
```

Default per-match delta bounds are -200 and +200. Preserve this order; do not perform percentage scaling only after averaging when the original scales each pair first. Clan progression updates the clan-owned subject through its own aggregation path, not each member's personal playlist XP. [S02, S13]

Individual performance is not an independent XP input. Gameplay statistics can still affect a mode's final score and therefore its standing. “XP depends on placement” does not permit a generic Slayer scoreboard to stand in for a flag, hill, ball, territory or role-based ending. [S13; N08–N10]

3.4 XP thresholds and hysteresis

Retail rank conversion is `0x74970`; recorded TU1.5 counterpart is `0x78790`. Find the ordinary threshold band. Retain the previous rank only when the ordinary result is exactly one raw rank lower and XP is strictly greater than that lower band's integer midpoint. At the midpoint itself, demote. A multi-band fall receives no one-step protection. [S02, S09, S13]

```
candidate = greatest raw index whose threshold is <= XP
if candidate == previous_rank - 1:
    midpoint = trunc((threshold[candidate] + threshold[candidate + 1]) / 2)
    if XP > midpoint:
        return previous_rank
return candidate
```

At the maximum-XP edge, repeated sentinel thresholds can yield raw 127. Presentation remains capped at level 50. For ordinary raw ranks 0–48, displayed level is raw+1; raw 49–127 share the final visible grouping. Do not truncate internal state to 49. [S02]

Retail boundary fixture: prior raw 10, lower band 2500–2999. XP 2751 retains 10; 2750 demotes to 9; 2499 falls to 8; 3000 is normally 10. TU1.5 fixture: prior raw 13, XP 1301 retains 13 while 1300 returns 12. [S09, S13]

3.5 Retail executable-default policy

Raw thresholds 0–49:

```
0,100,200,400,600,900,1200,1600,2000,2500,
3000,3500,4000,4500,5000,5500,6000,6500,7000,7500,
8000,8500,9000,9500,10000,10500,11000,11500,12000,12500,
13000,13500,14000,14500,15000,15500,16000,16500,17000,17500,
18100,18900,20000,21500,23500,26100,29400,33500,38500,44500
```

Raw thresholds 50–127 are `0x3fffffff`. For d=own−opponent, the initialized win table equals `clamp(100−10d,20,180)` and the loss table equals `−clamp(100+10d,20,180)`. These expressions describe these defaults, not every downloaded configuration. [S13]

Loss factors at raw 0–14 are `0,5,10,20,35,45,55,60,65,70,75,80,85,90,95`; raw 15–127 use 100. Win factors are 100 through raw 40, then `95,90,85,80,70,65,60,55,50,50` at raw 41–50, then zero at raw 51–127. Offsets are zero. [S13]

3.6 TU1.5 executable-default policy

Raw thresholds 0–49:

```
0,100,200,300,400,500,600,700,800,900,
1000,1100,1200,1400,1600,1800,2000,2250,2500,2750,
3000,3250,3500,3750,4000,4250,4500,4750,5000,5250,
5500,5750,6000,6250,6500,6750,7000,7250,7500,7750,
8000,8250,8500,8750,9000,9250,9500,9750,10000,10250
```

Again, indices 50–127 are `0x3fffffff`. Let k=min(abs(own−opponent),15). Favorite-win magnitudes A[k] are:

```
100,92,85,79,74,70,66,63,60,58,56,54,53,52,51,50
```

A favorite win is +A; a favorite loss is −(200−A). An underdog win is +(200−A); an underdog loss is −A. Apply the subject's factor per pair afterward. Equal rank has magnitude 100 in both directions. [S09]

Loss factors at raw 0–29 are:

```
0,2,5,7,10,15,20,27,35,40,45,50,55,57,60,
62,65,67,70,72,75,77,80,82,85,87,90,92,95,97
```

Raw 30–127 use 100. The win-factor tail, zero offsets and ±200 delta defaults remain as recorded above. These integer byte values resolve the later webpage's fractional and duplicated-row ambiguity for this executable's defaults. They do not authenticate the final-service download. Examples: raw 15 loses to raw 15 => -62; raw 41 defeats raw 40 => +87. [S09]

3.7 Ranked, casual and clan persistence

Hidden casual skill is a real per-playlist state, not a license for permanent skill-free matching. Public visibility and whether skill is stored are separate. Clan queues use a clan namespace. Display names and provider IDs must not become public competitive keys. Creating a new profile, renaming one or changing rank visibility must follow an explicit continuity/migration rule backed by the target epoch. [S11–S13, S19]

Do not add inactivity decay, placement matches, engagement optimization, forced win-rate targets, K/D-based XP or a new quit fine inside a compatibility profile unless actual evidence establishes that behavior for the chosen original reference. This is a preservation requirement, not a proof that every undocumented service-side administrative action has been recovered.

4. Search, gathering and candidate decisions

4.1 Preserve the search/gather architecture

The original client searches for advertised gathering sessions and can become a gathering owner itself. A persistent party and a temporary game/gathering session are not the same object. A centralized replacement can host these responsibilities, but ticket age, arbitrary player subsets and relay route scores are not automatically equivalent to the original session inputs. [S10, S11]

Keep admission, query construction, search expansion, candidate scoring, selection, whole-party join, failed-target memory, gathering growth and final team assignment separately testable. A correct score cannot repair a wrong candidate pool or premature role transition. The original server's result population and order remain a distinct unknown. [S10, S11]

4.2 Rank envelopes use extrema and an inverse table

The recorded retail forward table T has 128 entries. At index i, clamp i+offset to 127. Offsets are 5 for i=0–5; 6 for 6–15; 7 for 16–25; 8 for 26–30; 9 for 31–35; and 10 for 36–127. The inverse getter `0x1931A0` walks backward, not by a fixed subtraction. [S08; N02]

```
F(rank,T):
    i = rank
    while i > 0 and signed_byte(T[i-1]) >= rank:
        i -= 1
    return i

f = F(raw_maximum,T)
l = max(f,raw_minimum)
u = signed_byte(T[l])
R = max(l-f, u-raw_maximum)

at bounded radius r:
    minimum = max(l-r,f)
    maximum = min(raw_maximum+r,u)
```

Initial radius is zero. The advertised mapped minimum is `max(raw_minimum,F(raw_maximum,T))`. Missing-rank sentinels must be handled before valid indexing. Average rank remains a different candidate-score input. [S08]

Raw minimum… maximum	f	l	u	R	Initial bounds	Fully expanded
1…8	3	3	8	0	3…8	3…8
20…20	14	20	27	7	20…20	14…27
10…15	9	10	16	1	10…15	9…16
49…50	40	49	59	9	49…50	40…59

The project evidence distinguishes raw summary arrays from a separately computed effective/group floor and identifies a decompiled double-inverse mistake. Copying source that computes F(F(H)) when original controls establish F(H) would preserve the transcription bug, not the game. [S08]

4.3 Query exhaustion and casual fallback

The client requests up to 50 candidates per response and uses a recorded cumulative 150-result budget. Expansion follows completion of the current query and QoS work without a selectable target. Full-page continuation, radius exhaustion, failed-session memory and role choice must follow their actual branches. A final casual query may disable

rank eligibility bounds while retaining composite scoring. Ranked search does not acquire that casual escape branch. [S10, S11]

These rules are not equivalent to “broaden every N seconds” or “after 30 seconds ignore skill.” Older native documentation contains timed/lab policy descriptions; newer gathering integration must be selected by an explicit profile. Do not certify every entry point because one profile routes through the recovered search core. [S11, S12]

4.4 Composite score

Retail scorer `0x91000` evaluates one candidate gathering session for one searching party. With the recorded defaults, rank distance d contributes 100,85,70,55,40 for d=0…4 and zero outside that interval. This comes from `100−trunc((100−25)/(4+1))*d`, not interpolation to 25 at distance four. Exact remaining space equal to party size adds 50. Gathering age is clamped to 0…180 seconds and multiplied by 2.0. [S10]

The quality predicate adds 100 when either candidate quality value is below its threshold while both searching-party quality values meet their thresholds. The recorded thresholds are 262144 each. Do not reverse this apparently counterintuitive predicate. The inputs represent original measurement/estimate owners, not two arbitrary modern normalized percentages. [S10]

Language scoring uses the configured title-language matrix. For the recorded matrix, special indices are {1,6,7}: equal special indices add 100, equal ordinary indices add 0; differing indices subtract 100 if either is special and subtract 50 otherwise. Preserve numeric index provenance rather than inferring region or provider locale. [S10]

RTT uses the curve `100,80,50,0,-100,-200,-200,-200,-200,-200` at 100 ms intervals. Keep individual float32 operations before integer truncation:

```
t = clamp(f32(f32(rtt_ms) * f32(0.01)), 0, 9)
i = trunc(t)
if i < 9:
    contribution = trunc(f32(f32(curve[i]) +
                        f32(f32(t-f32(i)) *
                            f32(curve[i+1]-curve[i]))))
else:
    contribution = curve[9]
```

The recorded original result at 499 ms is -198, not the -199 produced by ideal-real interpolation followed by a single truncation. V4 reproduces the source-derived float32 contract; it does not rerun the Xbox instruction sequence. [S10; executed V4 contract checks]

4.5 Sorting, readiness and authentic inputs

The original sorts all 50 slots. Fully measured equal-score candidates have RTT/rank-distance/remaining-space tiebreaks; mixed-readiness comparison has asymmetric behavior. A stable tuple sort is not an approved substitute. Selection requires an eligible, fully measured candidate meeting the current threshold, with the original partial-readiness branch using the quarter-count test and strict elapsed-time comparison greater than 5000 ms. [S10]

The advertised age belongs to the gathering session, not a party ticket. Raw average belongs to the actual rated roster. Rank extrema belong to the mapped advertisement contract. Quality and RTT belong to their real network producers. A trusted measurement is still not an original-equivalent measurement merely because the server obtained it honestly. [S08, S10]

5. Team formation: original caller, kernel and integration correction

5.1 Hopper-derived options

The reconstructed caller `0x71730` obtains its parameters from the hopper getters. The getter bodies make the conditional contracts explicit. Their behavior must be replayed against the actual target executable before declaring the full caller closed. [N01, N02, N05]

Hopper type	Team path	Hidden rank class	Flag parties	Individual fallback
1	No; FFA	Yes	No	Not a team assignment branch
2	No; FFA	No	No	Not a team assignment branch
3	Yes	Yes	No	Allowed after preserved-party failure
4	Yes	No	Only when +0x604 is set	Not allowed by this caller
5	Yes; clan class	No	Yes	Not allowed by this caller

For team types, +0x5EC is team count, +0x5F0 minimum team size and +0x5F4 maximum team size in the reconstructed layout. Type 5's flagged-party interval comes from those team-size bounds. Type 4's enabled interval comes from +0x608/+0x60C. Type 3 maps the low byte of +0x5F8 to a count-imbalance allowance of 0 or 1; types 4/5 use the full field. Do not confuse count imbalance with the distinct significant-party-size spread rule. [N02]

5.2 Final assignment sequence

The traced final-gathering caller uses the raw weight array at summary +0xD4 and grouping array at +0x194. It does not pass the separate adjusted-player array merely because that array is adjacent. [N01; S08]

```
if hopper is FFA:
    assignment[i] = i
else:
    read team count, maximum size, flag enable/min/max and imbalance limit
    run original count balance with use_parties=true
    if feasible and spread <= hopper limit:
        run weighted balance with use_parties=true and optimize=true

    if assignment failed and the hopper permits the casual fallback:
        repeat count feasibility with use_parties=false
        if feasible and spread <= hopper limit:
            run weighted balance with use_parties=false and optimize=true

    preserve the original failure/transition behavior if both attempts fail
```

This closes a documentation gap in V3: the missing pieces are not merely “some party options.” They are identifiable getter contracts and ordered calls. It does not close the original-byte validation gap: `0x71730` remains marked `todo` upstream. [N01, N02, N05]

5.3 Greedy construction and local improvement

Kernel `0x91FC0` groups players by the selected party IDs, or individual seat index when preservation is disabled. Members are kept in descending weight order with original stable equal-weight behavior. Parties enter the pool in group-index order. The next party is selected by largest size, then largest total weight; exact ties retain traversal order. The destination team is selected by smallest player count, then smallest total weight, again retaining earlier exact ties. [N03, N04]

If the greedy result exceeds maximum team capacity, the kernel fails before optimization. It is not an exhaustive solver that must discover every theoretically feasible partition. When enabled, optimization considers whole-party swaps between teams, requires matching party-flag classes and rejects swaps increasing the pair's player-count difference. It prefers reduced global count spread, then strictly reduced cost. Iterate until no improvement remains. [N03, N04]

```
cost = 7 * (maximum team total - minimum team total)
     + 3 * sum over sorted member positions k:
               (maximum weight at k - minimum weight at k)

Missing member positions contribute 0.
Count-spread improvement has priority over cost improvement.
```

Traversal order, stable insertion, strict comparisons and signed arithmetic are part of the contract. The native kernel uses wrapping i32 arithmetic. Ordinary valid rank fixtures avoid overflow, but unrestricted “helpful” large weights are not an original validated input domain. There is no random selection among all globally minimal total-difference partitions in this kernel. [N03, N04]

5.4 Count and add-capacity helpers

`0x91E30` calls the kernel with zero weights and optimization disabled, then returns largest-minus-smallest team size. `0x91F00` considers additional seats, computes available capacity and checks `max(imbalance−open_slots,0) <= maximum_imbalance`. [N03, N04]

One source comment claims each extra seat is its own party, but the inspected loop assigns every extra seat the same next-party ID. The native kernel preserves the repeated ID. This is an example of why comments alone are not authoritative; retain an original-byte fixture that distinguishes the two behaviors. [N03, N04]

5.5 The known native divergence remains a required fix

At the pinned native commit, ordinary two-team matchmaking still uses a separate exhaustive partition optimizer minimizing total effective-rank difference, with random ties. More-than-two-team routing uses the native recovered kernel. Gathering finalization calls the same coordinator method, so it does not automatically avoid the two-team branch. The adapter also hardcodes flag settings to false/zero. [S03–S06]

For eight solo ranks `24,24,22,22,22,22,20,20`, the custom objective accepts both [24,24,20,20] versus [22,22,22,22] and matching [24,22,22,20] distributions. Both totals are 88 versus 88, but recovered costs are 24 and 0 respectively. The retained isolated counterexample finds 13 tied custom partitions:12 at cost 0 and one at cost 24. Its controlled tie chooser selects the latter. V4 reran this source-level counterexample successfully. [S03, S04; executed counterexample]

Required correction: establish the original caller contract, expose every required adapter option, implement the ordered feasibility/fallback path and route every applicable two-team profile through it. Do not merely change `>2` to `>=2`. Preserve clan admission and statistics separately; do not universalize “never split a party” where the original casual caller allows a fallback. [N01–N04; S06]

6. Map and variant selection: literal data and RNG ownership

6.1 Exact row identity matters

The inspected retail source defines five literal fallback tables containing 61 rows. The recorded complete fallback construction activates seven hopper blocks whose table expansion contains 116 row instances. V4 packages the literal triples with stable row IDs and duplicate rows intact. A selector for another block does not establish that the block was active. [N02; S09]

Table address	Literal rows	Selector block indices	Active fallback use
0x444FC0	14	0,13	Block 0
0x445068	24	1,4,6,14	Blocks 1,4,6
0x445188	12	3,5,15	Blocks 3,5
0x445218	5	11,12	None in recorded 0–6 active set
0x445258	6	2	Block 2

This JSON is not the original `multiplayer_hopper.dat`. It lacks full constructed settings and localized/header state and is not suitable for deployment as a complete historical catalog. Numeric preset IDs in these tables refer to 28 executable constructors, not the 39 UI built-ins. [N02; S09]

The recorded file layout has 16 headers of 0x614 bytes and 16 blocks of 0x15CB8 bytes, total 1,453,248 bytes. Each block contains up to 100 records; record stride is 0x37C, map ID at+4, weight at+0x248 and the full 304-byte settings buffer at+0x24C. A parser must validate structure, count, bounds and provenance without silently repairing a corrupted source. [S09]

6.2 Weighted choice and random step

Selector `0x192EB0` sets unavailable-map weights to zero. Available weights are clamped to 1…1000. It accumulates those weights in original row order, draws once, then selects the first cumulative sum at least as large as the draw. It copies the whole chosen variant, not just a name, mode or score limit. [N02]

The inspected random helper advances an unsigned 32-bit seed using the following LCG and returns its upper 16 bits. The weighted-range helper then scales that 16-bit draw. [N06]

```
seed = (seed * 0x0019660d + 0x3c6ef35f) modulo 2^32
u = seed >> 16

For the positive signed-short-safe total T:
    choice = 1 + ((u * T) >> 16)
    select the first row with cumulative_weight >= choice
```

Preserve duplicate rows as separate intervals. Preserve the installed-map predicate, initial seed, random-state owner and number/order of calls. A cryptographic random generator may be sensible for another service purpose, but replacing this draw inside an original profile changes seeded behavior. Python `random.choices` is not a drop-in parity implementation. [N02, N06]

6.3 Signed-short boundary and quantization

The reconstructed caller passes `(short)(total+1)` as the upper bound. That is a material numeric boundary. V4's executable reference model deliberately limits itself to totals at most 32766, where this upper bound stays positive and representable. The model rejects larger totals as an explicit harness safety restriction; that rejection is not claimed to be the original executable's behavior. Original-byte controls must establish narrowing, promotion and wrap behavior outside that domain. [N02]

Even in the safe domain, the draw has only 65536 possible values. For an ordered interval between cumulative weights A and B, its exact draw-bucket count is:

```
ceil(65536 * B / T) - ceil(65536 * A / T)
```

Thus ideal weight/T is a useful probability approximation, not necessarily the exact finite draw frequency. V4 enumerates all 65536 draw outputs for each of the five source tables and retains the resulting counts. That is a mathematical/source-contract check, not proof of historical seed distributions or Live rotation settings.

6.4 Full variant buffers remain mandatory

The fallback builder applies additional changes after invoking presets, including recorded flag and field overrides. A generic Slayer/CTF constructor plus a few settings is insufficient unless it produces the same full buffer. Keep the map-specific authoring, variant version, source buffer hash, normalization rules and engine type bound through selection, launch, result validation and replay. [S09, S13]

A published catalog may establish names and map associations without weights. An exact fallback may establish weights without proving Live deployed it unchanged. A recovered spreadsheet may establish editable settings without being a serialized title-storage file. These are complementary evidence classes, not interchangeable substitutes.

7. Every mode and preset: engine ownership through results

7.1 Registration identities

The TU1.5 generator reads a pointer cell, then the engine object, then its 51-method vtable. The following addresses are registration pointer cells, not function entrypoints. CTF and Assault share a registered object but remain distinct variant types. Unused types 5/6 are not new supported modes. [N07]

Family	Source type	TU1.5 registration cell	Built-in presets
Capture the Flag	1	0x47D174	4
Slayer	2	0x4822A0	9
Oddball	3	0x4822A4	6
King of the Hill	4	0x4822A8	5
Juggernaut	7	0x47A2F8	5
Territories	8	0x4822AC	5
Assault	9	0x47D174	5

The mode generator and project progress record describe substantially more than Slayer-only work. They report bounded callbacks, original/native component comparisons, real source-world startup, objective possession/scoring, zone behavior, role transitions, edited-limit endings and loopback transport checks. Those are valuable recorded controls, but they have explicit boundaries. V4 did not rerun them. [N07, N08]

7.2 Complete named built-in inventory

The actual result module maps the following 39 names to source engines. This is a concrete inventory for the built-ins, not an exhaustive list of every downloaded matchmaking variant. [N10]

Family	Named built-ins
Slayer	Slayer; Team Slayer; Rockets; Swords; Snipers; Phantoms; Team Phantoms; Elimination; Phantom Elim.
Oddball	Oddball; Rocketball; Swordball; Team Ball; Low Ball; Fiesta.
King	King; Team King; Phantom King; Crazy King; Team Crazy King.
Juggernaut	2 on 1; 3 on 1; Ninjanaut; Phantom Fodder; Dreadnaut.
CTF	Multi Flag CTF; CTF Classic; 1 Flag CTF; 1 Flag CTF Fast.
Assault	Multi Bomb; Single Bomb; Single Bomb Fast; Neutral Bomb; Blast Resort.
Territories	3 Plots; Land Grab; Gold Rush; Control Issues; Contention.

Notably, this built-in **Fiesta is Oddball**, not a general label authorizing a random-weapons Slayer implementation. Every historical variant must be identified from its authored engine/options, not a substring classifier. Likewise, a friendly display-name change must not detach the match from its actual options hash or statistics namespace. [N10]

7.3 Shared lifecycle integration

The generator identifies shared TU1.5 engine phases at 15EBA0,15F450,1503D0 and 13B990 and an authenticated fresh-player continuation. Preserve scheduling order, actual factory-owned player/unit creation, original lives, objective/role state and once-only result finalization. Supplying a completed score or fabricated current unit to a test is not equivalent to reaching that state through ordinary gameplay. [N07, N08]

The project record reports all 39 presets in several controlled cohorts. Some alter time or score limits, supply initial positions, or declare death-event inputs. Such tests can establish the specific lower-level contract they exercise. They do not replace unedited full-match and cross-machine acceptance. V4 requires the evidence ledger to record edited settings, synthetic inputs and replaced dependencies alongside each passing result. [N08]

7.4 Mode-specific acceptance

Family	Required natural outcomes and adverse cases
Slayer	FFA/team wins, true ties, finite-life exhaustion, score/time/round endings, suicides and retained departures.
CTF	Every evidenced flag variant; actual pickup/carry/drop/return/capture, side changes, overtime and final team placement.
Assault	Actual arming/planting/fuse/disarm or return rules, neutral/single/multi-bomb states, round sequencing and overtime.
Oddball	Real possession and periodic scoring, configured ball count, team/FFA differences, drop/departure and time-format results.
King	Occupancy, contest, score ownership, hill movement and exact rollover, multi-round endings and departures.
Territories	Neutralization versus capture intervals, owner changes, configured regions, round transition and final placement.
Juggernaut	Actual initial role selection, role transfer/reselection, scoring eligibility, cap/lives behavior and full final standings.

For every family, exercise launch through the correct hopper, disconnect and migration during meaningful state, original final standings, result commit, public/hidden rank refresh and party return. A component carrier test or loading-screen screenshot cannot complete this chain.

7.5 Postgame columns are not rank inputs

The seven-family postgame adapter records a distinct numeric projection for each engine. Retain the actual frozen source statistics and mode-selected formatting. These columns are presentation/summary contracts, not an invitation to add performance-weighted XP. [N09]

Source family	First summary field	Second summary field	Score format
CTF	stat 16	stat 15	Integer
Slayer	stat 13 / max(stat 9+1,1)	stat 12	Integer
Oddball	stat 26	stat 25	Original time

Source family	First summary field	Second summary field	Score format
King	stat 28	stat 27	Original time
Juggernaut	stat 38	stat 39	Integer
Territories	stat 43	stat 44	Original time
Assault	stat 21	stat 20	Integer

The original producer correspondence is recorded as retail 232D77 to TU1.5 23975C. PC/Vista carriers have different layouts and identity widths; using PC assets does not authorize substituting PC multiplayer rules or pretending a native opaque key is an original XUID. [N09, N13]

8. Historical playlists and the archival restoration register

8.1 Named checkpoints, not one timeless catalog

Maintain each checkpoint's publication date, announced activation date, confirmed deployment date, source capture date and configuration hash separately. Source pages can describe proposals that were delayed or changed. A closest archive snapshot is not evidence for the requested date. [S09, S18–S20; N18]

The seed inventory covers Rumble Pit, Double Team, Team Slayer, Team Skirmish, Team Training, Team Snipers, Team Hardcore, Team SWAT, Big Team Battle, Head to Head, Minor/Major Clan Match, later Clanmatch, 6v6 Team Battle, Rumble Armory, Team ActionSack, Rumble Preview, Team Preview, Rumble Training and the newly documented Team Carnage/HBO checkpoint. This remains an open inventory, not a claim that those are all activated names/revisions. [S09, S13, S19, S20; N15]

The May 9, 2007 release checkpoint identifies nine offerings: ranked Double Team, Team Slayer, Team Skirmish, Team Snipers and Team Hardcore; casual Rumble Pit, Team Training, Team SWAT and Big Team Battle. It does not by itself establish final 2010 numeric weights, full settings or every search/admission field. [S18]

8.2 Newly identified April 2006 settings asset

Bungie's April 28, 2006 update announces a downloadable spreadsheet of matchmaking game-variant settings for the April 2006 release. The archived link targets `gamevariants.xls`. V4 retrieved the announcement but could not retrieve the binary through the available download/archive routes. Its cells, sheets and values were not inspected. [N14]

The package preserves the exact lead and attempted access status. The next lawful recovery step is to obtain an archived copy, retain its original bytes/hash and source date, read every sheet and map each field to the original serializer. Do not infer that it contains weights, do not silently fix ambiguities, and do not relabel an April 2006 snapshot as final 2010.

8.3 Newly added Team Carnage/HBO deployment checkpoint

The preserved Bungie post dated August 22, 2006 identifies the deployed six-versus-six Team Carnage/HBO offering. It provides useful concrete settings: Sun King has a 180-second score target and 720-second match limit; Capture Madness uses 300-second attack rounds without ending on a capture. FAE was adapted to Slayer for matchmaking from an Oddball custom antecedent. A Chamb0 CTF time-field label contains a first-to count, so its duration must not be guessed. [N15]

These are partial configuration facts, not complete authored buffers, map associations or rotation weights. Preserve the source-label conflict as evidence. This checkpoint is now required in the inventory rather than silently omitted

because it was absent from an earlier seed list. The later weekly update corroborates the rotating community-playlist context. [N15, N16]

8.4 Preservation services and final-era recovery

Current Insignia material is useful for understanding a working modern compatibility service, but its community-updated playlists are not automatically original final-service configuration. Treat any comparison against it as a separate service/configuration reference, not as proof of what Bungie deployed in 2010. [N17]

Continue recovery from lawful user-owned caches, dated preservation holdings and authentic title-managed storage artifacts. Record byte hashes, game/build identifiers, file structure, collection context and evidence of deployment. A local fallback reconstructed by an executable is valuable even when localization is missing, but it remains a fallback artifact. [S09]

The final-service network and hopper blobs, full weighted catalog and exact server-side query/arbitration behavior were not recovered in this pass. The document therefore retains explicit blockers instead of presenting guessed defaults as closure.

9. Session lifecycle, host choice and running-world continuity

9.1 Distinct owners and states

Separate party leader, gathering host, gameplay simulation authority and result authority. The original can select/optimize a host before launch and can migrate authority during a running match. Those are different operations with different state and evidence requirements. Persistent party A, temporary game B and a modern authentication session must not collapse into one mutable object. [S11; N11, N12]

Gathering wait depends on population and original delay tables. At maximum valid count, the getter returns zero; outside the allowed count domain it returns INT_MAX; intermediate counts use the original table. The start-match owner then has its own host/readiness checks and recorded ten-second countdown. Reusing one arbitrary timer for all these stages is not parity. [N02, N11, N12]

9.2 Original host-control path

Owner	Original retail addresses	Contract
Graceful leave / proposal	614A0;61AC0	Select a successor from eligible current membership; selected peer must acknowledge before commit.
Host quality comparator	619B0	Multi-field comparator, not simply lowest RTT; some field units remain unresolved.
Abrupt host loss / election	62AB0;616E0;5E7F0;61EF0	Preserve source state eligibility, generation ordering, reachable/failed masks and election waits.
Reestablishment	61570;62550;61E00	New owner reestablishes channels/membership before stable host state; remove failed peers appropriately.
Running-world follow-through	83220;6AEE0;6AE20;83F00	Promote/demote actual world authority and remap surviving machines/players.

These are source/static-evidence mappings from the project research, not newly executed migration tests. The older note's statements about then-current native limitations must not be treated as a fresh audit of the latest continuity code. V4 requires a current implementation trace and live acceptance run. [N11]

9.3 Default timeout profile

Recorded retail defaults include 3000 ms proposal acknowledgment,10000 ms host-choice bound,10000 ms committed-candidate expiry,12000 ms reestablishment completion,5000 ms old-host retry delay,45000 ms hard election limit,4000 ms election silence/failure wait,6000 ms stable-candidate wait and 1000 ms peer-map broadcast interval. Changed candidate selection can permit a 500 ms send. These are configurable executable defaults, not universal historical service constants. [N11]

Preserve units, strictness, start/reset owner and generation of each clock. A timer value by itself is not enough to recreate a transition.

9.4 A host field change is not world migration

The original promotion chain repairs replication, scenario objects, engine state and result-recording ownership while retaining the existing match. The peer-election map is not a complete world checkpoint. A replacement transport must carry or preserve sufficient actual simulation state for the new host; it cannot manufacture continuity by changing a relay owner field. [N11]

All-mode acceptance must inspect live objective ownership, armed bombs/fuses, ball carriers, hills/territory state, Juggernaut role, player identities, scores, round clocks and retained quitters before and after transfer. Ordinary nonhost departure must not unconditionally terminate every survivor's game. Rejoin and backfill require their own historical evidence; a generic join API is not proof that original ranked matchmaking filled vacancies mid-match. [N11; engineering acceptance requirements]

10. Results, trust boundaries and modern infrastructure

10.1 Original standings and departed participants

Retain participants and their actual completed statistics through departure. The recorded original rules place finishing teams ahead of wholly departed teams and preserve the surviving team's result for a teammate who quit. FFA departure order, exhaustion, ties and variant-dependent comparison must come from the correct original producer, not a simple sorted score list. [S09, S13; N13]

Do not invent a fixed extra XP fine as part of the recovered rank function. Separately declared anti-abuse restrictions may change eligibility, but they must not be hidden inside original arithmetic. Actual original backend arbitration remains separate from these client/result semantics. [S09; N10, N13]

10.2 What the actual native result code does

The inspected validator checks completed-report flags, engine/team identity, admitted roster identities/names, team rows and placement bounds. Known built-in names map to the seven engine types; an unknown name falls back to an explicitly Slayer-oriented validation path. In a claimed original profile, every selected variant must instead be bound to its authenticated engine and full options, without relying on a name-based fallback. [N10]

This particular validator does not by itself prove that the full 304-byte options buffer is compared against the selected profile. Other admission layers may perform additional checks. V4 therefore requires a complete admission-to-result hash-binding audit; it does not assert that the entire service lacks such a check merely because one function does not show it. [N10]

The current commit condition requires confirmation from every originally admitted physical connection. It is not a majority vote and not automatically a surviving-player quorum. The schema preserves historical survivor-policy receipts without relabeling them as unanimous. The write records result, player/clan progression and history in one transaction and rejects a different digest for an already completed match. [N10]

10.3 The disconnection/arbitration gap is now explicit

When an admitted connection permanently disappears, test whether the remaining players can receive the outcome the original reference would have recorded. Do not assume an unreachable participant can later confirm. Conversely, do not replace unanimity with an insecure host-only or majority rule and label that original. Agreement among compromised clients is not independent simulation truth. [N10; S12]

The service needs a declared authority/dispute policy, trusted roster and epoch, bounded result receipt window, replay/quarantine behavior, idempotent commit and auditable before/after skill state. Those are modern requirements. Exact historical conflict resolution remains an open original-service question.

10.4 Recommended service boundaries

Component	Required responsibility
Identity / authorization	Verify Steam identity, issue opaque scoped sessions, enforce membership and single-clan rules.
Party / presence	Own roster generations, leader consent, local-seat/guest relationships and invitations.
Profile repository	Supply immutable executable/configuration/catalog/buffer identities with provenance.
Compatibility core	Run original-equivalent rank, search, candidate, assignment and draw decisions with explicit inputs.
Session / transport	Measure real routes, allocate secure connections, manage host generations and enforce roster authority.
Result authority / ledger	Validate evidence, resolve or quarantine disputes and apply XP once transactionally.
History / presentation	Publish privacy-appropriate projections without becoming competitive authority.

These are architectural recommendations, not a recovered Xbox Live database or endpoint schema. A modular service is preferable to microservices that introduce ordering and consistency changes before the original contracts are understood.

10.5 Release-bound API and persistence contract

An admission envelope should bind `match_id`, `party_generation`, `roster_hash`, `profile_id`, `executable_epoch`, `rank_policy_hash`, `hopper_hash`, `map_id`, `variant_buffer_hash`, `statistics_namespace`, `authority_epoch`, expiration and replay protection. Use authenticated opaque identities, not client-provided rank values. A result binds the same envelope plus retained placements, departed slots, evidence digest, frozen skills and once-only commit identity.

Keep party/gathering leases separate from competitive result transactions. Retries may return the same accepted result but must not reapply XP. A configuration switch during a match must not silently change its frozen rating tables. Do not manufacture XP from a nontrivial legacy display level when the underlying XP is unknown. These are modern consistency safeguards built around the preserved math. [S02, S12; N10]

10.6 Intentional deviations require a ledger

Relay-only routing changes RTT/reachability inputs. Centralized result preselection can change the candidate bank. New host scores can change authority. Different sanctions or disconnect policy can change eligibility or which games finalize. Dedicated servers alter the peer-host migration experience. Document each deviation's reason, scope, user-visible effect and tests; do not conflate it with a new XP model. [S04, S10–S12; N11]

11. Executed checks, acceptance gates and implementation order

11.1 What actually ran in V4

The included standard-library runner checks source-derived rank arithmetic, threshold/midpoint boundaries, raw/display separation, all raw extrema envelopes, candidate numeric edges, hopper-option classification, literal pool integrity, all 16-bit draw buckets for five tables, preset inventory and registry integrity. It reruns the retained isolated two-team counterexample. It requires no network access or game executable.

These are reference-model and data-integrity checks. They are not original-XBE differential tests. Expected arithmetic and data are derived from inspected/retained source evidence. Many assertions enumerate a finite numeric domain and are not independent gameplay scenarios. A large assertion count is not an accuracy percentage or a claim of complete mechanism coverage.

Check group	Executed assertions	Scope
rank aggregate	9	Signed division, tie divisor and clamps
rank tables	10	Transcribed table dimensions
rank pairs	131,074	Source-derived pair arithmetic over raw domain
rank boundaries	77,062	Threshold/midpoint reference comparisons
rank display	128	Internal/public separation
search inverse	128	All 128 forward/inverse relationships
search bounds	30,765	All raw extrema and legal radius bounds
candidate numeric	98	Float32 RTT, rank score and numeric gates
caller options	19	Hopper getter classifications
pool inventory	3	Literal counts and duplicate retention
weighted rng	327,733	Finite draw enumeration and safe-domain model
preset inventory	5	39-name/seven-family mapping
registry integrity	161	IDs, source references and explicit non-certification
counterexample	3	Retained isolated two-team mismatch

The recorded total is **567,198 executed assertions**, all passing. The release validator passes package structure but intentionally rejects complete certification because original/native execution receipts, historical completeness and deployment evidence are absent. Its strict invocation exits 2 rather than silently treating missing evidence as success.

```
python3 tests/contract_checks_v4.py
python3 tests/validate_parity_release.py
python3 tests/validate_parity_release.py --require-certification
```

The last command is expected to fail certification with the supplied research-only statuses. Do not toggle flags to make it pass. Attach valid evidence and complete the missing work.

11.2 Original-reference acceptance gates

Gate	Required completion evidence
G01 Target and provenance	Pinned original/native identities, dated profile scope, input units and caller layouts.
G02 Algorithm ownership	Every relevant registry entry has a verified caller, implementation route and reproducible receipt.

Gate	Required completion evidence
G03 Rank	All valid raw pairs/outcomes; absent/tied records; player/clan aggregation; thresholds, midpoint and maximum-XP edges.
G04 Search and candidates	All forward/inverse inputs; query exhaustion; original bank sort/readiness; authentic measurement fields and pool ordering boundary.
G05 Teams	Original caller options and kernel outputs for every original-valid topology, including two teams, guests, significant parties, failures and casual fallback.
G06 Catalog and draw	Complete full-buffer map/variant rows, duplicates, eligibility, weights, RNG state and numeric edge cases.
G07 Modes	Every required variant reaches its actual source engine and natural outcome; component-only or edited-limit runs are identified.
G08 Session continuity	Search/gather/launch/cancel, graceful/abrupt host changes, nonhost departures and retained running-world state.
G09 Results	Original standings, ties and departed records; actual finalization/disputes; exactly-once personal/clan XP and history projection.
G10 Historical profile	Authenticated dated inputs or a clearly bounded executable-default claim; complete included inventory without silent exclusions.
G11 Modern boundaries	Steam/security/relay changes isolated, intentional deviations recorded, custom/LAN/offline XP isolation proven.
G12 Deployed release	Tested binary/library/server/catalog/feature hashes match the actual running release and its smoke/acceptance traces.

Tests must state whether they compare bytes, decisions, timing transitions, serialization, statistical distributions or complete played outcomes. A replay should include mismatches and rejected cases, not only happy-path wins. Every skipped mandatory case remains a blocker.

11.3 Required fixture record

Each original/native comparison records the executable hash and address/version mapping, code/runner/fixture hashes, configuration bytes or digest, input structures, initial clock/RNG state, expected output, actual output, instruction substitutions, synthetic services, native build identity, environment and pass/failure result. Complete-match traces also record roster/authority epochs, actual mode actions and result commitments.

Do not leave the sole proof under an untracked `.local` path. Store lawful metadata, scripts and derived fixtures in durable evidence storage; keep copyrighted game executables/assets in the user's legitimate private reference environment rather than redistributing them with this document. [S07–S10]

11.4 Work sequence with objective exit criteria

Work package 1: prove and integrate the assignment caller. Recover the actual target caller/ABI, expose full adapter options and eliminate the two-team approximation in compatibility profiles. Exit only after the original and native full-caller outputs agree for the specified topology domain, including the retained counterexample and fallback failures.

Work package 2: close producer-to-consumer skill ownership. Trace raw, adjusted and persistent fields through parties, searches, assignments and results. Include separate machines, couch accounts, guests and clans. Exit with fixtures that distinguish each path, not one generalized mixed-party formula.

Work package 3: replay the complete search and draw pipeline. Preserve candidate-bank ordering, QoS readiness, failed sessions, radius progression and full selected settings. Add original controls around the signed-short draw limit and RNG call sequence. Exit with matching decision traces, not approximate match distributions alone.

Work package 4: qualify every required mode/profile entry. Start from genuine admission and ordinary actions, then exercise natural wins/ties, round ends, quits, migration, final results and requeue. Full authored settings and map revision remain bound throughout. A missing historical variant stays on the inventory.

Work package 5: settle historical inputs and service boundaries. Recover dated caches, spreadsheet/archive assets and original observations. Where original backend behavior remains unknown, isolate an explicitly named replacement policy without claiming the historical gap is closed.

Work package 6: certify a build and update the master ledger. Bind actual artifacts and deployment to all receipts. Profile completion and whole-program completion are separate decisions. Publish the precise tested domain and intentional deviations, not a universal percentage.

12. Original Algorithm Registry

12.1 Registry schema and maintenance

The companion JSON is the authoritative machine-readable index of this reference's 53 identified owners. It is not an assertion that all functions in the original game have been recovered. Each entry contains a stable ID, purpose, retail/TU1.5 addresses when established, address kind, input/output semantics, source IDs, native path, original harness path where known, matching status, local check group, remaining gaps and certification fields.

Registration pointer cells, function entrypoints, modern service policies and unrecovered backend contracts are explicitly different address kinds. TU1.5 addresses are not guessed by adding an offset to retail addresses. A source file's Git blob ID is not the original machine-code hash.

Every successful recovery or integration change must update its registry entry, fixtures and evidence receipt. The minimum review question is: “Which original owner, which inputs, which original configuration, which native output, and which actual comparison?” No entry becomes certified solely because its implementation name contains “original.”

12.2 Registry index

The following index is included for navigation. The full JSON contains the expanded input/output contract, paths and gap text. Original addresses marked R are retail; U are TU1.5. Empty numeric fields mean that the mapping was not independently established in this pass.

Algorithm ID / owner	Original address or kind	Contract / evidence boundary
ALG-RANK-INIT / Versioned rank configuration	R: 66330; U: 67400	128 thresholds, 255 exchange values per outcome, 128 percentages and offsets per outcome, delta limits. Source/report evidence; no V4 original-byte replay. [S02, S09, S13]
ALG-RANK-PAIR / Pairwise XP exchange	R: 746B0; U: 784D0	Signed pair delta. Source/report evidence; no V4 original-byte replay. [S02, S09, S13]
ALG-RANK-PLAYER / Player result aggregation	R: 74800; U: 78620	One bounded player XP delta. Source/report evidence; no V4 original-byte replay. [S02, S13]
ALG-RANK-CLAN / Clan-owned result aggregation	R: 74720; U: 78540	Clan XP delta, not personal playlist XP. Source/report evidence; no V4 original-byte replay. [S09, S13]
ALG-RANK-LEVEL / XP-to-raw-rank hysteresis	R: 74970; U: 78790	Raw rank 0..127. Source/report evidence; no V4 original-byte replay. [S02, S09, S13]
ALG-RANK-DISPLAY / Rank presentation mapping	asset and adapter contract	Displayed level/badge or hidden/NONE. Source/report evidence; no V4 original-byte replay. [S02, S12, N09]
ALG-PARTY-AVERAGE / Raw party average	R: 72540	Integer raw average and extrema. Source/report evidence; no V4 original-byte replay. [S08, S16, N05]
ALG-PARTY-SUMMARY / Gathering summary and separate effective array	R: 7E210	Raw array +0xD4, separate adjusted array, group IDs +0x194, extrema/average. Source/report evidence; no V4 original-byte replay. [S08, N01]
ALG-HOPPER-INVERSE / Inverse highest-matched-rank lookup	R: 1931A0	Lowest compatible raw rank F(rank). Source/report evidence; no V4 original-byte replay. [S08, N02]
ALG-HOPPER-GETTERS / Hopper classification and caller options	R: 1932C0, 193300, 193340, 193370, 1933A0, 1933D0, 193400, 193440, 193470, 193490, 1934B0, 1934D0	Counts, team geometry, imbalance, rank visibility, party flags and split fallback gate. Source/report evidence; no V4 original-byte replay. [N02]
ALG-SEARCH-INIT / Search envelope initialization	R: 90880	f=F(M), l=max(f,m), u=T[l], R=max(l-f,u-M), initial radius 0. Source/report evidence; no V4 original-byte replay. [S08, S10]

Algorithm ID / owner	Original address or kind	Contract / evidence boundary
ALG-SEARCH-QUERY / Bounded and unbounded query construction	R: 90B30, B4A90, 8FA80	Rank bounds,12 query positions; request up to 50 records via procedure 1. Source/report evidence; no V4 original-byte replay. [S08, S10]
ALG-SEARCH-EXPAND / Completion-driven query expansion	R: 90F70	Continue, widen, casual unbounded pass, or exhaust role. Source/report evidence; no V4 original-byte replay. [S11, S10]
ALG-SEARCH-SERVICE / Original server candidate population/order	unrecovered service contract	Candidate bank/order before client scoring. Original backend not recovered. [S10, S11]
ALG-CAND-ADMIT / Candidate admission	R: 90D90	Eligibility flag. Source/report evidence; no V4 original-byte replay. [S08, S10]
ALG-CAND-SCORE / Composite candidate scorer	R: 91000	Signed composite score. Source/report evidence; no V4 original-byte replay. [S10]
ALG-CAND-SORT / Complete candidate-bank ordering	R: 90F10, 13DA70	Original ordering with readiness-sensitive comparator. Source/report evidence; no V4 original-byte replay. [S10]
ALG-CAND-SELECT / QoS readiness and score threshold selection	R: 91400	Selected fully measured session or wait. Source/report evidence; no V4 original-byte replay. [S10]
ALG-QOS-INPUTS / Quality input producers	R: 7E100, 7F410, 64870, 658C0	Original quality pair in original units. Source/report evidence; no V4 original-byte replay. [S10]
ALG-TEAMS-CALLER / Final gathering assignment dispatcher	R: 71730	Assignment or continued/reset gathering. Source/report evidence; no V4 original-byte replay. [N01, N02, N05]
ALG-TEAMS-KERNEL / Greedy assignment and local party-swap optimization	R: 91FC0, 91A30, 91BD0, 91DD0, 91B90	Assignments or capacity failure. Source/report evidence; no V4 original-byte replay. [N03, N04]
ALG-TEAMS-COUNT / Party-preserving count balance	R: 91E30	Largest-minus-smallest player count or failure. Source/report evidence; no V4 original-byte replay. [N02, N03, N04]
ALG-TEAMS-CAPACITY / Partial-gathering add-capacity gate	R: 91F00	Boolean capacity/admission result. Source/report evidence; no V4 original-byte replay. [N03, N04]
ALG-TEAMS-ADAPTER / Native option propagation and two-team integration	native integration gap	Call to original-equivalent kernel. Known native divergence. [S04, S05, S06, N01, N04]
ALG-MAP-WEIGHTED / Ordered weighted map/variant selection	R: 192EB0	Selected map,304-byte variant, updated seed or failure. Source/report evidence; no V4 original-byte replay. [N02, N06, S09]
ALG-RNG-STEP / 32-bit LCG and high 16 extraction	R: 146240	Updated seed; upper 16-bit value. Source/report evidence; no V4 original-byte replay. [N06, N02]
ALG-HOPPER-FALLBACK / Executable fallback pool construction	R: 193010, 1940D0	Fallback 16-slot file with recorded active blocks 0..6. Source/report evidence; no V4 original-byte replay. [N02, S09]
ALG-VARIANT-PRESETS / Authored variant construction	R: 212F40, 19D220	Full 304-byte option buffer. Source/report evidence; no V4 original-byte replay. [S09, S13, N10]
ALG-COUNTDOWN-GATHER / Population-dependent gathering delay	R: 1931D0, 1939F0, 193A50	Wait/fill delay or INT_MAX invalid count. Source/report evidence; no V4 original-byte replay. [N02, S11]
ALG-COUNTDOWN-START / Separate start-match countdown	R: 72700	10-second countdown and launch transition. Source/report evidence; no V4 original-byte replay. [N12]
ALG-HOST-CHOICE / Host optimization comparator and proposal	R: 619B0, 61AC0	Chosen acknowledged successor / proposal / commit. Source/report evidence; no V4 original-byte replay. [N11]
ALG-HOST-ELECTION / Abrupt host-loss election	R: 62AB0, 616E0, 5E7F0, 61EF0	Election, candidate, promotion eligibility or timeout close. Source/report evidence; no V4 original-byte replay. [N11]

Algorithm ID / owner	Original address or kind	Contract / evidence boundary
ALG-HOST-REESTABLISH / Host reestablishment and membership repair	R: 61570, 62550, 61E00	Stable hosted session and survivors. Source/report evidence; no V4 original-byte replay. [N11]
ALG-WORLD-PROMOTE / Running simulation authority handoff	R: 83220, 6AEE0, 6AE20, 83F00	Continued world with repaired replication, engine and result ownership. Source/report evidence; no V4 original-byte replay. [N11]
ALG-RESULT-PLACE / Final standing and tie production	U: 2486F0, 248F20	Encoded ranks and final placements. Source/report evidence; no V4 original-byte replay. [N07, N08, N13]
ALG-RESULT-WIRE / Original results-transfer codec boundary	R: B2CC0, B2C50, B2C80, 197680, 197D80	Encoded/decoded result payload and validation. Source/report evidence; no V4 original-byte replay. [N13]
ALG-RESULT-PCR / Seven-family postgame numeric projection	R: 232D77; U: 23975C	Mode-specific numeric columns and time/integer formatting. Source/report evidence; no V4 original-byte replay. [N09]
ALG-RESULT-NATIVE / Native admitted-result validation and commit	modern native service policy	Single immutable transactional result/XP/history write. Source/report evidence; no V4 original-byte replay. [N10]
ALG-RESULT-ARBITRATION / Original Xbox Live result dispute policy	unrecovered service contract	Accepted/rejected authoritative competitive update. Original backend not recovered. [S12, S13, N10]
ALG-UI-FLOW / Search, roster concealment, cancel and postgame return	caller paths documented numeric mapping not rechecked	Correct UI state, concealed opponents before gate, postgame party return. Source/report evidence; no V4 original-byte replay. [N12]
ALG-ENGINE-FRAME / Shared TU1.5 engine lifecycle	U: 15EBA0, 15F450, 1503D0, 13B990	Mode update, round transitions, lives and result finalization. Source/report evidence; no V4 original-byte replay. [N07, N08]
ALG-ENGINE-NEWPLAYER / Authenticated player creation continuation	U: 14E737, 16E7F0, 14E73C	Source player/unit initialization. Source/report evidence; no V4 original-byte replay. [N07, N08]
ALG-MODE-CTF / CTF registered source engine	U: 47D174 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-SLAYER / Slayer registered source engine	U: 4822A0 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-ODDBALL / Oddball registered source engine	U: 4822A4 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-KING / King registered source engine	U: 4822A8 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-JUGGERNAUT / Juggernaut registered source engine	U: 47A2F8 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-TERRITORIES / Territories registered source engine	U: 4822AC (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-MODE-ASSAULT / Assault registered source engine	U: 47D174 (pointer cell)	Original engine state, scoring, role/objective behavior and final result inputs. Source/report evidence; no V4 original-byte replay. [N07, N08, N09, N10]
ALG-CLAN-IDENTITY / Clan subject and lineup constraints	identity and statistics contract	Admitted distinct opposing clan parties and clan-owned rank context. Source/report evidence; no V4 original-byte replay. [S09, S13]

Algorithm ID / owner	Original address or kind	Contract / evidence boundary
ALG-HISTORY-EPOCH / Historical catalog and statistics continuity	historical service input	Correct playlist profile and preserved/reset competitive state only where evidenced. Source/report evidence; no V4 original-byte replay. [S09, S19, N14, N15]
ALG-MODERN-TRANSPORT / Steam identity and secure network adapters	project requirement not original code	Authorized session operations without client-owned rank inputs. Source/report evidence; no V4 original-byte replay. [S11, S12, N10]
ALG-CERTIFICATION / Build-bound parity release decision	new validation control	Certified only for stated scope with zero unresolved mandatory gates. Source/report evidence; no V4 original-byte replay. [S07, N05]

12.3 How Codex must use this reference

Codex should read this master, the full registry, coverage register and gap register together. Reading the files supplies project context; it does not train model weights. The handoff contract requires source-first implementation, immutable evidence pins, explicit original/native receipts, no invented settings or tests, and registry updates with every correction.

The source-derived tests in this package are safe preflight checks. They are not substitutes for `tools/oracle_tu15_team_balance.py`, the original rank/candidate/mode harnesses or complete played-match acceptance in the project's legitimate reference environment. Preserve each harness's dependency and supplied-input limitations.

13. Gap closure register and final disposition

13.1 Documentation gaps closed or narrowed

V4 supplies an executable-address registry, traces the final assignment getter/flag/fallback sequence, records raw versus adjusted assignment ownership, transcribes literal fallback rows, specifies RNG/range/quantization semantics, maps all 39 built-ins to their exact seven source types, separates PCR presentation from XP, audits the actual result-confirmation branch and expands the historical source inventory.

These advances replace vague research objectives with identifiable contracts and tests. They do not change the source checkout or certify the implementation. The strongest still-demonstrated integration defect remains the ordinary two-team approximation; V4's counterexample rerun confirms that its documented source logic still differs.

13.2 Remaining mandatory closure work

Gap	Status / required closure
GAP-01: Two-team integration	Wire all applicable two-team profiles through authentic caller stages and native options; rerun original/native assignment traces and the V2 counterexample.
GAP-02: Caller and effective-skill provenance	Execute the original caller and producer chain, including remote parties, couch accounts, guests and clan subjects; record complete options and raw/effective values.
GAP-03: All-mode played-match acceptance	Play original-valid matches for every profile/variant route through natural endings, departures, result commit and requeue; no synthetic victory substitution.
GAP-04: Historical input recovery	Authenticate final-service network/hopper files, full variants and map weights; reconcile every dated activated playlist and mutation.
GAP-05: Original service candidate order	Recover original procedure or distinguishing ordered responses; until then label centralized preselection as a replacement service choice.

Gap	Status / required closure
GAP-06: Original arbitration and disconnected report owners	Obtain original behavior evidence; test native unreachable confirmer behavior and establish an explicitly named secure replacement where necessary.
GAP-07: Host continuity and QoS semantics	Inspect current native continuity code and prove all-mode world preservation with actual relay/direct measurements and host/departure permutations.
GAP-08: Weighted choice and full settings	Replay original RNG/caller context and signed-short boundary; export complete constructor/download settings; verify all selected map/variant bytes.
GAP-09: Reproducible original reference runs	Use legitimate pinned original executables, actual compiler/oracle dependencies and accessible immutable fixture artifacts. Record binary and instruction provenance.
GAP-10: Deployment identity	Verify runtime library, server, catalog and feature flags equal the tested build. Re-run acceptance against that release.
GAP-11: Full buffer and namespace binding	Prove that gameplay admission and result validation bind the selected 304-byte option hash, map revision, statistics namespace and rank policy across every entry point.
GAP-12: Operational documentation drift	Retire misleading legacy 30-second fallback/authentication/mode-gating statements only after checking actual profile dispatch; keep current UI/results evidence aligned.

An inaccessible asset is not proof that it no longer exists. A missing original-server implementation is not permission to invent its semantics. Continue recovery and experimental discrimination using lawful artifacts and explicit hypotheses, while keeping the interim product claims bounded.

13.3 Definition of done

The final result must cover the reconciled original historical inventory, route all applicable profiles through the correct original mechanisms, preserve every full authored map/variant row, produce correct natural mode outcomes and player/clan progression, survive the required session transitions, and bind that evidence to the deployed release. No required row may be “passed” because it was skipped, renamed out of scope or given a guessed default.

V4 is ready to guide implementation and further evidence recovery. It is not a claim that every historical-data gap is filled or that the full port is 1:1. Its explicit registry, executed checks and closure criteria make the remaining work testable instead of leaving Codex to improvise it.

14. Source register and citation discipline

Sources S01–S20 are retained from V3's register. Sources N01–N18 record the added V4 source inspection or archival research. The source manifest identifies which items were newly inspected, which remain reported controls, and which assets were not retrieved. Numbers and algorithms attributed to a report are not silently relabeled as fresh original-byte observations.

Repository links are immutable where pinned. An archived public statement can establish a dated design or configuration fact without establishing every field of a downloaded binary. Sources that contradict each other remain separate observations until the original version/input resolves the conflict.

S01. Repository baseline

Open source / pinned revision

Review class: retained v3 source register.

S02. Native rank engine and policy hashes

Open source / pinned revision

Review class: retained v3 source register.

S03. Recovered team kernel

Open source / pinned revision

Review class: retained v3 source register.

S04. Coordinator dispatch and host/game selection, reviewed source lines 400-640

Open source / pinned revision

Review class: retained v3 source register.

S05. Gathering finalization call to `_teams`

Open source / pinned revision

Review class: retained v3 source register.

S06. Native team adapter

Open source / pinned revision

Review class: retained v3 source register.

S07. Original TU1.5 team comparison harness

Open source / pinned revision

Review class: retained v3 source register.

S08. Rank-floor and search-envelope evidence

Open source / pinned revision

Review class: retained v3 source register.

S09. Historical configuration follow-up

Open source / pinned revision

Review class: retained v3 source register.

S10. Candidate scoring evidence

Open source / pinned revision

Review class: retained v3 source register.

S11. Native coordinator documentation, including legacy/new-profile overlap

Open source / pinned revision

Review class: retained v3 source register.

S12. Native progression and confirmation policy documentation

Open source / pinned revision

Review class: retained v3 source register.

S13. Original rank/playlist evidence

Open source / pinned revision

Review class: retained v3 source register.

S14. Bungie Weekly Update, April 15, 2005, preserved by HBO

Open source / pinned revision

Review class: retained v3 source register.

S15. Bungie Halo 2 Stats Overview, preserved by Podtacular

Open source / pinned revision

Review class: retained v3 source register.

S16. Effective-rank and party-average helpers, reviewed lines 190-320

Open source / pinned revision

Review class: retained v3 source register.

S17. Original Xbox Halo 2 human manual, printed pp. 22-25 (PDF pages 13-14), seven game types and custom-game rank separation; checked October 28, 2026

Open source / pinned revision

Review class: retained v3 source register.

S18. Original Bungie release announcement, May 9, 2007, preserved blog; nine playlists and visibility, checked October 28, 2026

Open source / pinned revision

Review class: retained v3 source register.

S19. Original Bungie Weekly Update, December 2, 2005, preserved article; announced catalog changes, rank continuity and stale-catalog behavior, checked October 28, 2026

Open source / pinned revision

Review class: retained v3 source register.

S20. Official Bungie playlist update post, July 12, 2005; Preview playlists and map gates, checked October 28, 2026

Open source / pinned revision

Review class: retained v3 source register.

N01. Retail final-gathering assignment caller

Open source / pinned revision

Review class: source inspected v4.

N02. Hopper getter contracts and weighted map selector

Open source / pinned revision

Review class: source inspected v4.

N03. Independent reconstructed team-balancing source

Open source / pinned revision

Review class: source inspected v4.

N04. Native Rust balancing implementation

Open source / pinned revision

Review class: source inspected v4.

N05. Per-function match-status inventory

Open source / pinned revision

Review class: matching search excerpt inspected v4.

N06. 32-bit RNG step / 16-bit draw helper

Open source / pinned revision

Review class: rng definition search excerpt and header inspected v4.

N07. TU1.5 engine registration and shared lifecycle recovery

Open source / pinned revision

Review class: source inspected v4.

N08. All-mode operational evidence and explicit limitations

Open source / pinned revision

Review class: source inspected v4.

N09. Seven-family postgame adapter

Open source / pinned revision

Review class: source inspected v4.

N10. Actual result validation / 39-preset map / unanimous commit

Open source / pinned revision

Review class: source inspected v4.

N11. Original host-departure and world-promotion research

Open source / pinned revision

Review class: recorded static original analysis inspected v4 not replayed.

N12. Matchmaking frontend / countdown / return flow

Open source / pinned revision

Review class: recorded original controls and adapter description inspected v4.

N13. PC versus Xbox postgame placement and transfer boundaries

Open source / pinned revision

Review class: recorded controls inspected v4 with platform boundaries.

N14. Bungie Weekly Update, April 28, 2006: variant spreadsheet announcement

Open source / pinned revision

Review class: primary text retrieved v4 spreadsheet not retrieved.

N15. Bungie Team Carnage / HBO deployment post, August 22, 2006

Open source / pinned revision

Review class: archived primary deployment text retrieved v4.

N16. Bungie Weekly Update, September 1, 2006

Open source / pinned revision

Review class: primary text retrieved v4.

N17. Insignia Halo 2 service page

Open source / pinned revision

Review class: current preservation service not historical configuration.

N18. Bungie Weekly Update, July 28, 2006: proposal

Open source / pinned revision

Review class: primary proposal not deployment evidence.

Executed local evidence: `tests/results_v4.json` and `tests/release_gate_result_v4.json`, generated by the included scripts. The result JSON expressly records that original XBE execution, compiled Rust execution, complete native-service execution and deployed-service inspection did not occur. The package manifest contains delivered-file hashes.

15. Gap recovery: what is missing and where to find it

V4.1 recovery edition | Added October 28, 2026

The entire preceding V4 reference is retained. This new section expands its twelve GAP records into practical recovery instructions. It does not replace the formulas, source pins, evidence labels, acceptance gates or original audit findings.

Some gaps are missing client-side analysis, some are missing historical data, some belong to the retired service, and some are unfinished implementation or testing. These require different evidence. Decompiling the whole client is not a substitute for recovering a later download or running a complete match.

15.1 The four places to investigate

Evidence location	What to look for	What it cannot establish alone
Original executable and title updates	Calculations, defaults, loaders, callers, session state and reporting paths.	The contents of a later downloaded configuration or private server code.
Original assets and retained downloads	Built-in variants; captured network/hopper files; dated complete map/variant settings.	When a file was deployed, unless collection and service provenance also survive.
Original service records and disclosures	Title-specific queries, ordered responses, registration/report behavior and dispute outcomes.	That a generic platform design or a modern replacement was Halo 2's production policy.
Reference environment and actual port	Original/native comparisons, full mode/playlist flows and tested release hashes.	Historical settings that were never supplied to the test.

15.2 Evidence boundary for this edition

The source basis is the uploaded V4 master and the supplied Missing-Evidence Recovery Addendum. Sections 1-14 remain the historical record of what V4 inspected and tested. This appendix organizes that evidence and proposes recovery work; it makes no fresh claim that repositories, caches, original binaries or deployments were inspected.

The addendum narrowed GAP-04, GAP-05 and GAP-06 with a workbook lead and platform disclosures. It did not recover authenticated final-2010 files or Halo 2's server implementation. No gap is marked closed here. [Recovery addendum, Result and What remains unresolved]

Navigation: The next page maps all twelve gaps. Sections 15.4-15.11 identify owners, files, experiments and closure evidence. Sections 15.12-15.13 give safe acquisition procedures and source links.

15.3 Recovery map for all twelve open gaps

Use the existing GAP identifiers when assigning work or updating the registry. “Where to look” is a recovery target, not a claim that the required artifact has already been found. All status remains OPEN. [V4 §13.2]

Gap	Primary work / first place to look	Close only with
01 Two-team integration	Native coordinator, gathering dispatcher and team adapter; original final-assignment caller. See §15.4.	Actual routed caller and original/native outputs agree.
02 Caller / effective skill	Original raw/effective/XP producers and consumers; rank-floor evidence and result producer. See §15.4.	Versioned caller traces for remote, local, guest and clan cases.
03 All-mode acceptance	Mode owners, generated fixtures, natural-match tests and actual matchmaking entry points. See §15.5.	Full applicable variant/profile traces, not synthetic victories.
04 Historical inputs	Retained TMS downloads, original assets, dated archives and the April 2006 workbook. See §15.6-15.7.	Authenticated profile data, complete entries and defensible dates.
05 Server candidate order	Client query/result boundary plus title-specific server definitions or captured responses. See §15.8.	Evidence distinguishing filtering, ordering, truncation and ties.
06 Arbitration	Client registration/report path plus original decision records; native result commit. See §15.9.	Title-specific missing/conflicting-report behavior or a labeled replacement.
07 Host / QoS continuity	Original host-election and world-promotion owners; native continuity and route adapters. See §15.10.	All-mode authority transfer and authentic measurement semantics.
08 Weighted choice / settings	Original selector, RNG owner, preset/fallback constructors and selected full buffer. See §15.7.	Original draw/call-sequence replay and complete settings identity.
09 Reproducible originals	Legitimate reference executables, actual generators and retained fixtures/receipts. See §15.5.	Repeatable original/native runs with input and build hashes.
10 Deployment identity	Running client/library/server, active catalog and feature configuration. See §15.11.	Proof the deployed bundle equals the tested bundle.
11 Buffer / namespace binding	Admission, room state, gameplay handoff, result validator and persistence. See §15.9.	One selected full-buffer/profile identity survives every transition.
12 Documentation drift	Active route/configuration selection, mode gates and identity provider. See §15.11.	Documentation reconciled to observed implementation and receipts.

A repository path is an owner locator, not proof of execution. A generated fixture is not an original binary. A passing shared kernel does not certify every caller or playlist. Keep these distinctions in the registry and release gates. [V4 §2.2, §11.2-11.3, §12.1]

15.4 GAP-01 and GAP-02: recover and connect client logic

GAP-01 | Original team assignment is not the current two-team route

What is missing: The actual two-team compatibility path must use the correct original caller sequence and options, not merely expose a recovered kernel. V4 identifies a custom two-team partition optimizer and an adapter that hardcodes party flags. [V4 §5.5; S04-S06]

Where to look: In the native repository, follow multiplayer/master_server/matchmaking.py, gathering_sessions.py and team_balance.py into rust/online_protocol/src/team_balance.rs. In the pinned independent decompilation, inspect src/unknown_071730.cpp, src/unknown_1932c0.cpp and src/team_balancing.cpp. [S03-S07; N01-N04]

Original owners: Retail 0x71730 dispatches final assignment; 0x91FC0 assigns teams; 0x91E30 checks count balance; 0x91F00 checks additional capacity. The hopper getters determine flags, count limits and the permitted casual fallback. These retail addresses are not TU1.5 addresses. [V4 §5.1-5.4]

Recovery procedure: Trace the selected hopper into the count-feasibility call, weighted call and permitted retry. Preserve group order, raw weights, party flags, optimization, capacity and failure behavior. Reproduce the retained eight-solo counterexample through the real dispatcher, then compare the complete original and native caller outputs.

Closure evidence: Caller/ABI record, original code identity, updated adapter contract, actual dispatch trace and original/native results for FFA, two-team, valid multi-team and clan exceptions. A one-character branch edit or standalone kernel test is insufficient. [V4 GAP-01]

GAP-02 | Prove raw, adjusted and persistent skill ownership

What is missing: A verified producer-to-consumer chain for each grouping case. Do not assume that party average, candidate bounds, assignment weight and XP comparison skill are the same value. [V4 §3.1, §4.2, §5.2]

Where to look: Start with multiplayer/docs/halo2-rank-floor-evidence.md and online-rank-playlist-evidence.md, the native playlist_skill.py, and the original routines they identify. Retail 0x72540 aggregates raw ranks; 0x7E210 produces the summary and separate adjusted array; 0x71730 consumes assignment inputs. Follow the result producer into 0x746B0/0x74800 without substituting a capped display level. [S02, S08, S13, S16; N01]

Recovery procedure: Record each field at producer, publication and consumer boundaries. Exercise equal/mixed ranks, multiple remote machines, multiple signed-in local seats, guests and clan subjects. Include a case distinguishing F(H) from the reported double-inverse transcription error F(F(H)).

Closure evidence: Complete versioned traces with grouping, raw/effective/previous rank, XP and relevant buffers. Confirm the target instructions where reconstructed source remains unmatched. The public mixed-party example alone does not prove every caller branch. [V4 GAP-02]

15.5 GAP-03 and GAP-09: build a usable reference environment

GAP-09 | Reproducibility starts with the actual reference inputs

Where to look: Use the developer's legitimate local copies of the retail and chosen title-update executables, original assets, oracle dependencies, generated fixtures and test receipts. V4 records hashes and source pins, but explicitly says it did not possess or independently execute the original binaries. [V4 §2.1, §2.4, §11.3]

Start with: tools/oracle_tu15_team_balance.py; the rank and candidate harnesses referenced by S13 and S10; tools/generate_multiplayer_engines_tu15.py; docs/GAME_MODE_PROGRESS.md; and the fixture locations declared by those tools. A path mentioned in research must be checked on the actual workstation before treating it as an available artifact. [S07, S10, S13; N07, N08]

Recover or regenerate: Inventory each required input, verify its hash, preserve the original script/dependency version and reproduce the reference receipt. Regenerate a missing fixture from the authenticated original input. Do not generate expected values from the native implementation under test or replace an unavailable original call with a convenient return value.

Closure evidence: A repeatable invocation, pinned inputs, runner/compiler/environment identity, reference outputs, native outputs, complete mismatch log and an explicit account of synthetic inputs or substituted services. Private original game files do not need to be distributed with the public document. [V4 GAP-09]

GAP-03 | All modes and playlists must work end to end

The seven families are CTF, Slayer, Oddball, King, Juggernaut, Territories and Assault. V4 identifies 39 built-in presets, but downloaded historical variants extend beyond them. Existing bounded engine and shortened-round records remain useful; they are not full historical-profile acceptance. [V4 §7.1-7.4; N07-N10]

Where to look: Read the actual mode registration/generator, game-mode progress evidence, online-pcr-families-tu15.md and match_results.py. Follow a selected playlist entry into its exact source engine and through the same result/XP path used by other profiles. [N07-N10]

```
admission > search/gather > formation > selected full variant
> natural mode outcome > retained departures > final standings
> accepted result > personal/hidden/clan XP > party return
```

Required runs: For every applicable profile/variant route, exercise natural wins and ties, time/round endings, local and remote seats, participant departure, meaningful host migration and postgame/requeue. Include a no-matchmaking-XP check for custom/LAN/offline play. Record edited limits or declared events as component-only evidence.

Closure evidence: Original-reference and native traces identify the same settings, state inputs and observed outputs. Record missing, skipped and failed cases; do not count one successful Slayer match or one engine callback as all-mode completion. [V4 GAP-03; §11.2]

15.6 GAP-04A: recover dated network and hopper inputs

What is missing: Authenticated historical network and hopper configurations, especially the final-service target, plus enough dated evidence to establish which settings were active. The code can expose defaults and the loader without containing a later downloaded file. [V4 §1.3, §8.4; S09]

What the executable can and cannot provide

Inspect the original configuration initializer, file-registration/download/cache path, version acceptance and fallback behavior. Determine whether each observed byte came from an embedded default, a loaded historical download, or a replacement service. A memory capture taken after loading a modern configuration is not an original-service artifact.

Target	First places to investigate	Useful evidence
network_configuration.dat	Lawful preserved title-storage downloads; read-only old-console/reference copies; local captures and configuration research S09/S13.	Original bytes, parsing/acceptance result, executable identity and collection/deployment provenance.
multiplayer_hopper.dat	The same preserved downloads/copies; the original loader and fallback builder in src/unknown_1932c0.cpp; dated Bungie catalogs.	Complete headers, entries, weights and full variant buffers, with a dated source.
Actual in-memory configuration	Original client running against a controlled, clearly identified reference input.	A trace of default initialization, successful file load and the exact settings consumed by search/rank code.

Acquisition and authentication requirements

Before connecting an old installation to an update or replacement service, make a read-only preservation copy when feasible. Work on the copy and preserve collection context. The presence or location of these files on any specific console has not been established by this reference; use the original file owner to locate them rather than assuming a universal cache path.

Keep original filename or container entry, byte length, SHA-256, collector/source, collection time, title/build context and the evidence for the claimed service date. Distinguish a disk modification timestamp, an archive capture date and a demonstrated deployment date. A hash proves identity, not historical origin.

V4 records a retail fallback hopper layout of 16 headers and 16 blocks, total 1,453,248 bytes, with complete 304-byte settings per entry. Use it as a version-specific structural check, not as proof that every later file must have exactly that layout or that a matching-size file is authentic. [V4 §6.1; S09]

Close GAP-04A when: The claimed profile has authenticated inputs, its parser/load trace is understood, the full selected catalog is decoded without silently repaired fields, and the dating claim is supported. Until then, retain “retail defaults,” “TU1.5 defaults” or a clearly named reconstructed profile, not “final-2010 identical.”

15.7 GAP-04B and GAP-08: full variants, weights and draws

Separate three different content sources

Source	Where to look	Boundary
Built-in variants	Original game assets and the appropriate preset constructors; mode/preset generator and recorded controls.	39 menu presets are not the same inventory as the 28 executable constructors used by fallback rows.
Executable fallback pools	Retail 0x193010 / 0x1940D0 and src/unknown_1932c0.cpp; S09 construction evidence.	The 61 literal rows and 116 active row instances are fallback evidence, not final Live settings.
Downloaded matchmaking variants	Authenticated hopper files, preserved full settings, dated releases and the 2006 spreadsheet lead.	A variant name or settings list alone does not prove full serialized bytes, map overrides or weights.

Workbook recovery lead: April 28, 2006

The supplied recovery addendum reports recognizable but damaged extraction from gamevariants.xls, linked by Bungie's April 28, 2006 update. The complete binary, worksheet coordinates and reliable cell values were not acquired. V4's earlier “not retrieved” statement remains a record of that pass; the later extraction is a lead, not a recovered workbook. [R01-R02]

Obtain original bytes losslessly, hash them, retain the announcement/source date and inspect every worksheet. Preserve blanks, formulas, source labels and unknowns. Map settings into the original serializer and compare complete output buffers. Do not assume the workbook contains weights, and do not relabel its 2006 snapshot as final 2010.

GAP-08 | The selector needs original state, not just weights

Where to look: Retail selector 0x192EB0, preset 0x212F40 and its constructors, fallback 0x193010/0x1940D0, RNG step 0x146240 and the state/call sites feeding the selector. The independent source locators are src/unknown_1932c0.cpp and include/unknown_0259d0.h. [V4 §6; N02, N06; S09, S13]

Required comparisons: Available-map filtering; clamped weights; original row and duplicate order; draw boundaries; random-state ownership and consumption; the signed-short upper-bound edge; and the entire chosen 304-byte settings buffer after constructor and map-specific overrides. Preserve the original failure path and output writes.

Closure evidence: For each included profile, an authenticated map/version/variant/weight manifest plus original/native selection traces for identical inputs and random state. Every chosen byte must remain bound into launch and results under GAP-11. Matching long-run proportions alone does not prove seeded decision parity. [V4 GAP-08]

15.8 GAP-05: the server selects the candidate bank

What is missing: Halo 2's title-specific server query behavior before client scoring: session filtering, ordering, result truncation and ties. Reproducing the local candidate score does not establish which sessions the original service returned. [V4 §4.1, §12 ALG-SEARCH-SERVICE]

What can be recovered from the client

Trace retail 0x8FA80 (the search wrapper), 0xB4A90 (parameter construction), 0x90B30 (query stage) and the returned-record parser into candidate admission/scoring. V4 records procedure index 1, twelve query positions and a requested maximum of fifty candidates. Preserve parameter types, null/absent behavior, result limits, returned order and error/retry semantics. [V4 registry; S08, S10]

Start with the source paths cited by halo2-candidate-scoring-evidence.md and the pinned query/search owners, rather than assuming semantic names from a newer Xbox API. This can establish the boundary contract and what the client does with a particular response. It does not recover unseen server-side code.

Where the remaining service evidence could be found

Investigate authorized original title-service/query definitions, preservation holdings with original request/response captures, dated session observations and records supplied by original developers or service maintainers. These are search targets; this edition does not claim any holder has the missing definition.

The recovery addendum identifies two platform disclosures: US20040002384A1 documents title-specific XOnlineMatchSearch procedures and the API contract; US7634569B2 describes a proximity-ordering design. Use them to formulate competing hypotheses, not to declare that Halo 2 used every described branch. [R03-R04]

How to discriminate between hypotheses

For each original observation, retain the query inputs, observable candidate population, server response order, response size, time and client version. Look for cases that separate proximity order, rank order, age order, database/tie order and truncation behavior. A result consistent with several hypotheses does not select one.

A controlled replacement server can feed fixed response banks to the original client and reveal its admission, sort and join decisions. That is a client-behavior experiment. It is not an observation of the retired server's own selection process.

Closure evidence: The original procedure or a stated, sufficiently discriminating body of title-specific original evidence that fixes the claimed behavior. Until then, describe native server preselection and ordering as a replacement policy, while still testing the recovered client decision core. [V4 GAP-05]

15.9 GAP-06 and GAP-11: accepted results and bound settings

GAP-06 | Client reporting is not the whole arbitration service

Where to look in code: Start with online-rank-playlist-evidence.md and online-postgame-evidence.md, then follow original registration/report/result paths. Retail B2CC0/B2C50/B2C80 and 197680/197D80 identify a result-transfer boundary, not the entire private service decision. Compare the actual native multiplayer/master_server/match_results.py and its callers. [S13; N10, N13; V4 registry]

V4 records confirmation from every originally admitted physical connection as the native commit condition. An unreachable owner, one console with multiple local seats, late reports and conflicting reports must be tested explicitly. A schema or policy name does not prove what the original service did. [V4 §10.2-10.3]

Where to seek service evidence: Authorized original registration/report traces, accepted/rejected outcomes, service definitions or preserved operational records. The public arbitration disclosure US7584154B1 and trust disclosure US7367888B1 are documented design leads. Their deadline/dispute concepts and trust counters are not authenticated Halo 2 production parameters. Report credibility is not competitive XP. [R05-R06]

Closure evidence: Title-specific outcomes for missing, late, duplicate, conflicting and disconnected reports, including participant ownership and the final competitive write. Where that evidence is absent, implement a separately named secure replacement and document its differences; do not call it recovered original arbitration. [V4 GAP-06]

GAP-11 | Keep one selected profile identity through every stage

What is missing: An end-to-end proof that the actual full variant, map revision, rank policy and statistics namespace remain bound from selection into gameplay and result validation. V4 warns that one validator's checks do not establish the complete upstream chain. [V4 §10.2, §10.5]

Where to look: The catalog loader, gathering selection, proposal/admission records, room/gameplay handoff, result validator, playlist_skill.py and clan result persistence. Begin with gathering_sessions.py, matchmaking.py and match_results.py, then follow the actual callers and stored fields. [S02, S04, S05; N10]

Required checks: Keep separate display names and authoritative variant IDs. Compare the complete 304-byte options or their authenticated digest. Reject changed engine/team/options, wrong map revision, wrong statistics subject, changed policy or stale admission. A historic objective variant with an unfamiliar name must not silently enter a Slayer fallback.

Closure evidence: A trace and regression tests showing the selected immutable envelope at every consumer, including restart/retry and duplicate commit. Record where an upstream layer provides a check instead of assuming its absence from one function proves it is absent everywhere. [V4 GAP-11]

15.10 GAP-07: host migration and QoS need real state

What is missing: A current native continuity audit and original-reference tests demonstrating that the running world and all required modes survive the appropriate host/departure cases. Original host selection also consumes specific network inputs; a relay score is not equivalent merely because it is measured honestly. [V4 §4.5, §9; S10; N11]

Original owners to trace

Owner	Retail locators retained by V4	What to observe
Quality inputs	7E100, 7F410, 64870, 658C0	Input origin, units, availability, sample age and propagation into score/host selection.
Graceful selection / proposal	614A0, 619B0, 61AC0	Eligible members, generations, comparison and selected-successor acknowledgment.
Abrupt loss / election	62AB0, 616E0, 5E7F0, 61EF0	State eligibility, known/failed/reachable masks, candidate changes and timeout ownership.
Reestablishment	61570, 62550, 61E00	Channel/membership restoration, survivor handling and transition to stable host state.
Running-world handoff	83220, 6AEE0, 6AE20, 83F00	Actual authority promotion/demotion, object/engine/recording repair and participant identity.

Where to look: multiplayer/docs/halo2-host-migration-evidence.md and halo2-candidate-scoring-evidence.md provide the initial source map. Trace current runtime/transport owners from the actual checked-out branch; older notes describing fixed native roles are not proof that those limitations still apply to a later implementation. [N11; S10; V4 §9.2]

Required all-mode continuity experiments

Record the world immediately before loss, during election/reestablishment and after authority changes. Include a carried flag/ball, an armed bomb with its fuse, an active hill or contested territory, Juggernaut role, player/unit identities, score, round timer and retained departed records. The surviving match must continue through its real mode owner, not a newly constructed replacement game.

Distinguish pregame host optimization, graceful gameplay-host departure, abrupt loss and ordinary nonhost departure. A changed relay host field proves neither world promotion nor preservation. Rejoin and mid-match backfill require independent historical evidence; a generic join facility is not enough. [V4 §9.1-9.4]

Closure evidence: Versioned original/native state-transition traces, complete input-owner mapping and actual relay/direct acceptance tests with declared transport differences. Preserve configurable default timing as a profile input rather than an eternal service constant. [V4 GAP-07]

15.11 GAP-10 and GAP-12: prove the release and update its claims

GAP-10 | The running build is an evidence source

What is missing: Verification that the actual executable, native library, server, catalog, full settings and feature switches equal the artifacts that passed acceptance. Source inspection and a successful development build do not establish deployment identity. [V4 GAP-10; §11.2 G12]

Where to look: The real launch/deployment configuration, executable/library/container identities, catalog/variant files, selected profile IDs, enabled route flags and retained runtime receipts. Obtain these from the authorized running environment, not from a readme that names a presumed default.

Required work: Freeze a release manifest, verify the active processes against it, and run production-equivalent smoke/acceptance flows. Include configuration change during an outstanding ticket or match, restart during result commit and rollback of the compatible bundle. Record any stale library, divergent catalog or route override.

Closure evidence: A build-bound manifest, observed runtime identities and repeatable acceptance receipts. This is operational verification; no amount of original game decompilation proves which library your server currently loads.

GAP-12 | Documentation must follow verified dispatch

Where to look: multiplayer/docs/matchmaking.md, ranked-progression.md, online-matchmaking-flow-tu15.md, the real catalog selection and active authentication/queue paths. Preserve the original source-pin record while updating current operational descriptions separately. [S11, S12; N12]

V4 highlights overlapping descriptions of timed casual fallback, recovered gathering behavior, identity providers and mode support. Resolve them by tracing the selected profile and observing the running route. Do not delete an old limitation until a newer source pin and appropriate test demonstrate the replacement behavior.

Steam identity, one-clan membership and modern safeguards remain project requirements. Their presence in this specification is not independent evidence that the current executable or deployment satisfies them. [V4 §1.4, §10.4]

Closure evidence: Documentation tied to verified current routes, versions and receipts; superseded notes labeled historical; compatibility claims limited to the observed tested domain. Keep genuine unsupported cases visible rather than dropping them from the inventory.

Completion order without waiting on every archive

Run two tracks in parallel: client/integration verification (GAP-01/02/03/07/08/09/11) and historical/service recovery (GAP-04/05/06). Then bind the evidence to a deployed profile (GAP-10) and reconcile operational documentation (GAP-12). Dependency order may vary, but a passing kernel cannot close an unauthenticated historical profile.

15.12 Practical acquisition and evidence checklist

Read-only discovery on a legitimate reference copy

The following commands are proposed workstation steps, not actions performed by this document update. Set ROOT to a read-only copy of the relevant installation/import/preservation collection. Filenames can be absent, renamed or packaged in a container; a negative filename search is not proof of absence.

```
ROOT="/path/to/read-only-reference-copy"
find "$ROOT" -type f \
 \( -iname "network_configuration.dat" \
   -o -iname "multiplayer_hopper.dat" \
   -o -iname "gamevariants.xls" \
   -o -iname "default.xbe" \
   -o -iname "mainmenu.map" \
 \) -print

# Hash a selected artifact without changing it:
sha256sum -- "/path/to/selected/artifact"
```

For source locators, use the declared repository root and search for both filenames and owner addresses. Record git rev-parse HEAD and local modifications before comparing against V4 pins. Follow the original loader when packaged storage obscures the filename. Never boot/update the sole preservation copy just to see whether a file appears.

Minimum evidence receipt for each candidate artifact

```
gap_id; artifact_class; original_filename_or_container_entry
collection_source; collection_time_utc; source_build_or_title
byte_length; sha256; read_only_copy_location
claimed_service_epoch; evidence_supporting_that_epoch
parser_or_runner_commit; input_configuration_hashes
expected_outputs; actual_outputs; mismatches; test_scope
synthetic_inputs_or_services; instruction_replacements
status = acquired | parsed | provenance_pending | replayed
closed = false until the applicable closure criterion is met
```

Keep “not yet acquired,” “acquired but unparsed,” “parsed but undated,” “source-traced,” “original/native replayed” and “deployment-verified” separate. Each answers a different question. Preserve failed cases, unchanged original files and acquisition receipts; never repair source data in place.

Access and privacy boundary

Share scripts, non-sensitive manifests and permitted derived evidence. Keep original game executables/assets in the legitimate private reference environment. Remove authentication secrets, session keys and personal identifiers from captures intended for sharing. A public GitHub omission may reflect local/private storage; it does not establish that the developer lacks the reference input.

15.13 Source locators and recovery handoff

The original S01-S20 and N01-N18 links remain in Section 14. Use those immutable source pins for the retained audit. The labels below point to the two repositories already named by V4; they do not identify new inspected revisions.

Native repository: TheRunaway5/Halo-2-Decompiled, V4 audit snapshot

Independent reference: kirklandsig/halo2-decompiled, V4 source snapshot

Recovery addendum sources: leads and their limits

R01. Bungie Weekly Update, April 28, 2006

Announcement supporting a dated variant-settings workbook lead.

R02. gamevariants.xls target

Prior extraction was damaged; no authenticated full workbook or reliable cell map is supplied.

R03. US20040002384A1: session discovery

Platform search/procedure contract, not Halo 2's procedure-1 body.

R04. US7634569B2: proximity matchmaking

A possible ordering design, not demonstrated Halo 2 production configuration.

R05. US7584154B1: result arbitration

A registration/report/dispute design, not title-specific deployed policy.

R06. US7367888B1: player trust

Report-credibility lead; no recovered trust equation or new competitive rating rule.

R07. Halo Archive

Operator-described historical match corpus; raw XP and full variant data access not established.

R08. Insignia Halo 2

Modern preservation service; authenticate any claimed original-era configuration separately.

Source status is retained from the supplied recovery addendum. URLs are investigation targets, not packaged source binaries, fresh availability checks or proof of a holder's possession of the missing artifact.

Required handoff to the implementation agent

Read the retained master, algorithm registry, coverage register and this recovery map together. For each GAP, produce the identified artifact or executed comparison, record its provenance, update the affected algorithm/coverage rows and retain regression failures. Do not close a gap from a hypothesis, a generic patent, a modern-service match or a test that never ran.

The final target is unchanged: all required original-valid modes and historical playlist profiles must pass the complete claimed mechanism on the identified release. This edition clarifies where to look and what proof is required; it does not declare any remaining gap solved.